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High-Dimensional Statistical Inference and Variable Selection Using Sufficient Dimension Association

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ABSTRACT

Simultaneous variable selection and statistical inference is challenging in high-dimensional data analysis. Most existing post-selection inference methods require explicitly specified regression models, which are often linear, as well as sparsity in the regression model. The performance of such procedures can be poor under either misspecified nonlinear models or a violation of the sparsity assumption. In this article, we propose a sufficient dimension association (SDA) technique that measures the association between each predictor and the response variable conditioning on other predictors in the high-dimensional setting. Our proposed SDA method requires neither a specific form of regression model nor sparsity in the regression. Alternatively, our method assumes normalized or Gaussian-distributed predictors with a Markov blanket property. We propose an estimator for the SDA and prove asymptotic properties for the estimator. We construct three types of test statistics for the SDA and propose a multiple testing procedure to control the false discovery rate. Extensive simulation studies have been conducted to show the validity and superiority of our SDA method. Gene expression data from the Alzheimer Disease Neuroimaging Initiative are used to demonstrate a real application. Supplementary materials for this article are available online, including a standardized description of the materials available for reproducing the work.

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1. Introduction

As we venture further into the era of big data, the proliferation of expansive datasets presents a novel array of analytical complexities. High-dimensionality remains one of the most important complexities, where thousands of predictors are commonly available for only hundreds or even tens of samples. For example, the Alzheimer Disease Neuroimaging Initiative (ADNI) study collects longitudinal clinical, brain imaging, and gene expression data to support Alzheimer's Disease research. The ADNI Gene Expression Profile, a single dataset from ADNI, contains microarray data of 49,386 probes from a total of 745 different individuals. A critical task for such high-dimensional data analysis is to identify important features (e.g., probes) that are associated with the outcome of interest.

A prevalent strategy for variable selection in high-dimensional data analysis is to use regularization-based regression methods, such as LASSO (Tibshirani 1996), SCAD (Fan and Li 2001), MCP (Zhang 2010), and many others. These methods generally prescribe explicitly defined regression models and assume sparsity in the regression models. Besides variable selection, post-selection inference has emerged as a significant research direction in the past decade. The goal of post-selection inference is to derive valid statistical inference by accounting for the uncertainty inherent in the selection

(Kuchibhotla, Kolassa, and Kuffner 2022). For example, Lee et al. (2016), Tibshirani et al. (2018), and McCloskey (2023) have explored the conditional selective inference approach, where the inference can be made by conditioning on the selection procedure. The efficacy of the conditional selective inference method is contingent upon the performance of the focused selection procedure.

Sufficient dimension reduction (SDR, Cook 1998) is a dimension reduction method for variable selection in low-dimensional settings. Under this framework, dimension reduction is achieved by assuming that there exist low-rank subspaces of the original covariate space, or dimension reduction spaces, such that the outcome is independent of the covariates when conditioning on the projection of the covariates onto these subspaces. The sliced inverse regression (SIR) proposed by Li (1991) is the most popular approach for SDR. In high-dimensional settings, the sparsity of eigenvectors in dimension reduction spaces is often assumed, with research primarily focusing on consistently estimating the central subspace, that is, the smallest dimension reduction space (Ni, Cook, and Tsai 2005; Lin, Zhao, and Liu 2018, 2019; Lin et al. 2021). However, studies on inference for each covariate in high-dimensional settings are still limited. Zhu, Miao, and Peng (2006) investigated the limiting distribution of SIR in fixed dimensions, while Zhao and Xing

(2022) extended this to diverging dimensions and introduced a mirror statistic approach for false discovery rate (FDR) control based on data splitting.

This article introduces a novel statistical inference method for high-dimensional settings using SDR. We first examine the necessary conditions for a predictor to be part of the Markov blanket (Candes et al. 2018), which is the minimal set of variables encapsulating the dependency between outcome and covariates. According to the derived conditions, we propose to make inference and select variables based on a measure named sufficient dimension association (SDA). Using the assumption of multivariate normality and sparsity of the precision matrix, we propose a LASSO-based estimator for the SDA, which can be used to test the significance of each covariate separately. Contrary to most existing SDR methods, our proposed method does not require the central subspace to be consistently estimated. To test each covariate's membership in the Markov blanket, we construct a simple χ^2 statistic and two other statistics based on the Kolmogorov-Smirnov (KS) and Cramér-von-Mises (CvM) principles. A multiple testing procedure has been proposed to control the FDR.

The proposed SDA method does not require any explicitly specified regression model as opposed to most existing post-selection inference methods. This method is practically simple to understand and implement. Despite the many sophisticated variable selection methods that have been developed for high-dimensional data, the concept of univariate association testing is still popular in scientific applications. The SDA enjoys such simplicity as it is merely a (conditional) association measure for each univariate predictor.

The proposed SDA has a tie to the concept of partial correlation in the literature. A partial correlation refers to the correlation between two random variables after adjusting for the effect of a set of controlling variables (Baba, Shibata, and Sibuya 2004). Assuming a joint Gaussian distribution for the response and the covariates, Bühlmann, Kalisch, and Maathuis (2010) proposed the partial trustfulness and a PC-simple algorithm. Li, Liu, and Lou (2017) extended the method to elliptical linear regression models. Alabiso and Shang (2023) studied the partial faithfulness for high-dimensional linear mixed-effects models and Liu, Lou, and Li (2018) considered variable selection for partial linear models. For variable screening, Xia and Li (2021) proposed a copula-based partial correlation measure, Lu and Lin (2020) considered the conditional distance correlation measure, and Huang, Deb, and Sen (2022) developed the kernel partial correlation measure. These existing methods focus on the variable selection consistency and the sure screening property, while statistical inference remains unclear. On the inference side for high-dimensional linear models, Gong, Zhang, and Liu (2018) developed a test based on the maximum of a sequence of partial correlations, and Hemerik, Thoresen, and Finos (2021) considered permutation tests based on partial or semipartial correlations.

The rest of the article is organized as follows. Section 2.1 outlines notations and assumptions used in this article. Section 2.2 introduces the measure for sufficient dimension association. Section 2.3 discusses the relationship between the proposed SDA and the partial correlation measure. Section 2.4 reviews the sliced inverse regression. Section 3.1 presents the proposed

estimator for SDA, while Section 3.2 delves into its theoretical properties. Sections 3.3 and 3.4 introduce standard error estimation and hypothesis testing methods, respectively. The finite sample performance of the proposed method is evaluated in Section 4 through extensive simulations. In Section 5, the method is applied to gene expression data from the ADNI study, focusing on identifying genes linked to Alzheimer's disease. We conclude the article with a thorough discussion in Section 6.

2. Sufficient Dimension Association

2.1. Assumptions and Notations

Throughout this article, the superscript 0 is used to represent the true value of a given parameter. For any vector $\mathbf{a} \in \mathbb{R}^n$, a_i denote the i th coordinate of $\mathbf{a}$ for any $i \in \{1, \dots, p\}$, $\mathbf{a}(\mathbb{I})$ denote the subvector of $\mathbf{a}$ with coordinates of $\mathbb{I}$ for any $\mathbb{I} \subset \{1, \dots, p\}$, and $\mathbf{a}_{-i}$ is used to denote the subvector of $\mathbf{a}$ excluding the i th coordinate. Similarly, for any $n \times p$ matrix $\mathbf{A}$, A_{ji} denotes the (j, i) th element of $\mathbf{A}$ for any $j \in \{1, \dots, n\}$ and $i \in \{1, \dots, p\}$, $\mathbf{A}(\mathbb{J}, \mathbb{I})$ denotes the submatrix of $\mathbf{A}$ with rows of $\mathbb{J}$ and columns of $\mathbb{I}$ for any $\mathbb{J} \subset \{1, \dots, n\}$ and $\mathbb{I} \subset \{1, \dots, p\}$, and $\mathbf{A}_{-j, -i}$ is used to denote the submatrix of $\mathbf{A}$ excluding the j th row and the i th column. For any set $\mathbb{S}$, $|\mathbb{S}|$ denotes the cardinality of $\mathbb{S}$.

Let $\mathbf{X} = (X_1, \dots, X_p)^\top$ be a p -dimensional vector of predictors and Y is the response. Then $\mathbf{y} = (Y_1, \dots, Y_n)^\top$ denotes the vector of n response observations and $\mathbf{x} = (\mathbf{x}_1, \dots, \mathbf{x}_n)^\top$ denotes the corresponding covariate matrix, where $\mathbf{x}_j = (X_{j,1}, \dots, X_{j,p})^\top$ is the covariates vector for subject j . We assume that $\mathbf{X}$ is normalized with mean zero and covariance matrix Σ . We consider the semiparametric model $Y = f(\mathbf{b}_1^\top \mathbf{X}, \dots, \mathbf{b}_d^\top \mathbf{X}, \epsilon) = f(\mathbf{B}^\top \mathbf{X}, \epsilon)$, where $f(\cdot)$ is an arbitrary unknown function, $\mathbf{b}_1, \dots, \mathbf{b}_d$ are unknown vectors, $\mathbf{B} = (\mathbf{b}_1, \dots, \mathbf{b}_d)$, and ϵ is independent of $\mathbf{X}$ with mean zero. The linear space spanned by $\mathbf{B}$, denoted as $\text{col}(\mathbf{B})$, is called a dimension reduction space. The intersection of all dimension reduction spaces, denoted as $\mathbb{S}_{Y|X}$, is called the central subspace for the regression of Y on $\mathbf{X}$ (Li 1991; Cook 1996). The $\mathbb{S}_{Y|X}$ is, by definition, unique and can capture all information of Y given $\mathbf{X}$. By assuming an elliptical distribution for $\mathbf{X}$ and the *linearity condition*:

(C1) Linearity condition: $E(\mathbf{a}^\top \mathbf{X} | \mathbf{b}_1^\top \mathbf{X}, \dots, \mathbf{b}_d^\top \mathbf{X})$ is a linear combination of $\mathbf{b}_1^\top \mathbf{X}, \dots, \mathbf{b}_d^\top \mathbf{X}$ for every $\mathbf{a} \in \mathbb{R}^p$,

Li (1991) showed that $\Sigma \mathbb{S}_{Y|X} = \text{col}(\Lambda)$, where $\Lambda := E\{\text{var}(\mathbf{X} | \mathbf{Y})\}$. In this article, we further assume that $\mathbf{X}$ follows multivariate Gaussian, that is, $\mathbf{X} \sim N(0, \Theta^{-1})$, where $\Theta = \Sigma^{-1}$ is the precision matrix of $\mathbf{X}$.

For high-dimensional settings, we also assume that the dependence between Y and $\mathbf{X}$ can be characterized through a Markov blanket $\mathbf{X}(\mathbb{A})$ (Candes et al. 2018), that is,

$$Y \perp\!\!\!\perp \mathbf{X} | \mathbf{X}(\mathbb{A}), \quad (1)$$

where $\mathbb{A} \subset \{1, \dots, p\}$ is the minimal index set satisfying (1). In this context, we aim to make inference about whether predictor i belongs to the set $\mathbb{A}$, which translates to the following hypothesis testing problem:

$$H_0 : Y \perp\!\!\!\perp X_i | \mathbf{X}_{-i} \text{ versus } H_1 : Y \not\perp\!\!\!\perp X_i | \mathbf{X}_{-i}. \quad (2)$$

The connection between the Markov blanket assumption (1) and the conditional association test (2) is discussed in the supplemental material. Unlike other high-dimensional approaches such as knockoffs (Barber and Candès 2015) and selective inference (Taylor and Tibshirani 2018), we do not require a sparse regression model, that is, $|\mathcal{A}| \ll p$. Rather, we assume that the precision matrix Θ is sparse, that is, $I_i = |I_i| \ll p$ where $I_i = \{i : \Theta_{i,i} \neq 0\}$ for every $i \in \mathcal{P}$. Note that the predictor X_i is conditionally independent with X_i if $\Theta_{i,i} = 0$. The sparsity of conditional dependency in covariates is commonly assumed in the SDR literature, examples include but not limit to (Tan et al. 2018; Lin, Zhao, and Liu 2018; Pircalabelu and Artemiou 2021).

Under the linearity condition (C1) and letting $\delta(Y) = \Theta E(X|Y)$ (Li 1991, Theorem 3.1), we have $\delta(Y) \in \mathcal{S}_{Y|X}$, which also implies that $E\{g(Y)\delta(Y)\} = \Theta \text{cov}\{X, g(Y)\} \in \mathcal{S}_{Y|X}$, where $g(\cdot) \in \mathcal{F}$ and $\mathcal{F}$ is a sequence of transformation functions of Y . Thus, with a sequence of transformation functions $\{g_h(\cdot)\}_{h=1}^H$, let $\beta_h := \Theta \text{cov}\{X, g_h(Y)\}$ for every $h \in \mathcal{H}$, where $\mathcal{H} = \{1, \dots, H\}$. Hence, we have $\text{Span}(\beta_1, \dots, \beta_H) \subseteq \mathcal{S}_{Y|X}$. Similar to Cook and Ni (2006), Wu and Li (2011), and Zhao and Xing (2022), we also assume the *coverage condition*:

(C2) Coverage condition: $\text{Span}(\beta_1, \dots, \beta_H) = \mathcal{S}_{Y|X}$ when $H > d$.

2.2. Measure of Sufficient Dimension Association

To establish an association between each predictor X_i and the outcome variable Y while controlling all other predictors, we start from the dependence structure within X . Let β_{hi} be the i th element of β_h and we have $\beta_{hi} = \theta_i^\top \text{cov}\{X, g_h(Y)\}$, where θ_i is the i th column of the precision matrix Θ . Due to the property of multivariate Gaussian distribution, the conditional distribution of X_i given all other predictors X_{-i} follows a Gaussian distribution,

$$X_i|X_{-i} \sim N(-\sigma_i^2 \theta_{i,-i}^\top X_{-i}, \sigma_i^2), \quad (3)$$

where $\sigma_i^2 = \theta_{ii}^{-1} > 0$ and $\theta_{i,-i}$ denotes θ_i excluding the i th element. Thus, let $\zeta_i = -\sigma_i^2 \theta_{i,-i}$ and then we can write (3) as a linear regression

$$X_i = \zeta_i^\top X_{-i} + Z_i, \quad (4)$$

where $Z_i \sim N(0, \sigma_i^2)$. Since we have assumed that the precision matrix Θ is sparse, consequently, the induced $\theta_{i,-i}$ and ζ_i are also sparse, with $\theta_{ij} = \zeta_{ij} = 0$ if $j \notin I_i$.

Recall the assumption that the dependence between Y and X is determined by a Markov blanket (1). The problem is to identify the Markov blanket $X(\mathcal{A})$ and make inference for each individual predictor X_i . The following proposition reveals the link between the sufficient dimension method and the membership of X_i to $X(\mathcal{A})$.

Proposition 1. Assume that conditions (C1) and (C2) hold. Then $i \in \mathcal{A}^c$ if $\text{cov}(Z_i, g_h(Y)) = 0$ for all $h \in \mathcal{H}$, and $i \in \mathcal{A}$ if there exists $h \in \mathcal{H}$ such that $\text{cov}(Z_i, g_h(Y)) \neq 0$.

Proof. The coverage condition suggests that all values in $\mathcal{S}_{Y|X}$ can be written as a linear combination of $\beta_1, \dots, \beta_H$, which implies that $i \in \mathcal{A}^c$ if and only if $\rho_{1i}^0 = \dots = \rho_{Hi}^0 = 0$. Since

$\sigma_i^2 > 0$, the scaled parameter $\mathbf{v}_i := \sigma_i^2 \beta_i$ inherits the property of β_i . From (3) and (4), denote v_{hi} the h th component of $\mathbf{v}_i$, we have

$$v_{hi} = E(\sigma_i^2 \theta_i^\top X g_h(Y)) = E(Z_i g_h(Y)) = \text{cov}(Z_i, g_h(Y)).$$

This is because $\theta_i^\top X = \theta_{ii} X_i + \theta_{i,-i}^\top X_{-i} = \sigma_i^{-2} (X_i + \sigma_i^2 \theta_{i,-i}^\top X_{-i}) = \sigma_i^{-2} (X_i - \zeta_i^\top X_{-i}) = \sigma_i^{-2} Z_i$. The proof is complete. $\square$

Proposition 1 suggests that to test whether X_i belongs to the Markov blanket $X(\mathcal{A})$, that is, the conditional dependence between Y and X_i given X_{-i} , we can test the marginal association between Y and Z_i through the covariance $\text{cov}(Z_i, g_h(Y))$.

We introduce the concept *sufficient dimension association*, as defined in the proof of Proposition 1: $v_{hi} = \text{cov}(Z_i, g_h(Y))$, for a sequence of transformation functions $g_h(\cdot)$, $h \in \mathcal{H}$. The SDA sequence of v_{hi} is a measure of conditional association between an individual predictor X_i and the outcome Y given all other predictors. For this association measure, we make no assumption about the regression function f . However, we need to test a sequence of H hypotheses, that is, $H_0 : \text{cov}(Z_i, g_h(Y)) = 0$ versus $H_1 : \text{cov}(Z_i, g_h(Y)) \neq 0$, to satisfy the coverage condition. As a special case, for linear regression models, the SDA can reduce to the correlation between Z_i and Y .

2.3. Relationship to the Partial Correlation

The SDA measure is related to the partial or semipartial correlation measures (Cohen et al. 2013). The partial correlation is defined as

$$\rho_{YX_i|X_{-i}} = \text{corr}\{Y - E(Y|X_{-i}), X_i - E(X_i|X_{-i})\}, \quad (5)$$

and the semipartial correlation is defined as

$$\rho_{Y(X_i|X_{-i})} = \text{corr}\{Y, X_i - E(X_i|X_{-i})\}, \quad (6)$$

for which linear models are typically assumed for the conditional expectations $E(Y|X_{-i})$ and $E(X_i|X_{-i})$. The partial correlation is often used to measure the conditional dependence for Gaussian linear models, where the multivariate Gaussian assumption implies linearity for $E(Y|X_{-i})$ and $E(X_i|X_{-i})$. However, when the model is nonlinear, the partial correlation disagrees with the conditional correlation in general, known as the inconsistency (Baba, Shibata, and Sibuya 2004; Vargha, Bergman, and Delaney 2013).

The inconsistency in nonlinear models suggests that a single linear measure is insufficient to capture the nonlinear conditional dependence between Y and X_i . To address this issue, existing approaches relax or modify the linearity assumptions on $E(Y|X_{-i})$ and $E(X_i|X_{-i})$ (Huang 2010; Shah and Peters 2020), or replace the Pearson's correlation in (5) with nonlinear alternatives, such as rank-based (Xia and Li 2021), distance-based (Wang et al. 2015; Lu and Lin 2020), or kernel-based (Huang, Deb, and Sen 2022) correlation measures. Indeed, Shah and Peters (2020) argues that there does not exist a uniformly valid conditional independence test for all problems. Methods relying on the nonparametric regression (Huang 2010; Wang et al. 2015)

are not feasible in high-dimensional settings. Semiparametric approaches, such as Huang, Deb, and Sen (2022), are more scalable for high-dimensional variable selection, but valid inference remains challenging. Methods using parametric approaches to adjust for the confounding effect $\mathbf{X}_{-i}$, such as Xia and Li (2021), may still be vulnerable to model misspecification.

Our proposed SDA measure has a unique advantage that the nonlinear relationship between Y and $\mathbf{X}_{-i}$ can be unspecified while requiring that $\mathbf{X}$ be multivariate Gaussian, which is equivalent to a linear model assumption for $E(\mathbf{X}_i|\mathbf{X}_{-i})$. The key to address the inconsistency and capture the nonlinear association between Y and Z_i is to use a sequence of covariance measures $\text{cov}(Z_i, g_h(Y))$, which is distinct from existing approaches in the literature. Using the theory of SDR, our proposed method is flexible for a wide class of models in high-dimensional settings. In a later section, we show that our testing procedure only requires fitting a single high-dimensional linear model, which is more computationally efficient than some tests based on high-dimensional partial correlation measures, for instance, the permutation test proposed by Hemerik, Thoresen, and Finos (2021). Lastly, we note that for linear regression models, with $H = 1$ and $g_1(Y) = Y$, the SDA is equivalent to the semipartial correlation specified in (6).

2.4. Sliced Inverse Regression

Different choices of $\{g_h(\cdot)\}_{h=1}^H$ have been proposed in the literature, examples of which can be found in Yin and Cook (2002), Cook and Ni (2006), and Wu and Li (2011). In this article, we focus on the sliced inverse regression (SIR) method proposed by Li (1991), where the response variable Y is discretized into H slices, that is, $g_h(y) = I(y \in \mathcal{I}_h)$, where $\mathcal{I}_h$ is the set of all possible values for the h th slice. When Y is a categorical random variable or only takes on a few values, each category or each unique value naturally defines a slice. When Y is a continuous variable, the range of Y is divided into H slices based on a nondecreasing sequence $\{a_h\}_{h=0}^H$ with $a_0 \leq \min(Y)$, $a_H \geq \max(Y)$, and $\mathcal{I}_h = \{y : y \in (a_{h-1}, a_h)\}$.

3. Statistical Inference

3.1. Target of Estimation

We are interested in estimating the SDA sequence $\{v_{hi}, h \in \mathcal{H}\}$ and testing the sequence of hypotheses for SDA. Using the SIR technique, v_{hi} can be expressed as $v_{hi} = \text{cov}\{Z_i, I(Y \in \mathcal{I}_h)\} = E\{I(Y \in \mathcal{I}_h)Z_i\}$. We propose an estimator for the SDA in the following form

$$\hat{v}_{hi} = \frac{1}{n} \sum_{j=1}^n I(Y_j \in \mathcal{I}_h)(X_{ij} - \hat{\boldsymbol{\xi}}_i^\top \mathbf{X}_{-ij}),$$

where $\hat{\boldsymbol{\xi}}_i$ is an estimator of $\boldsymbol{\xi}_i$ from the linear model (4). The number H will be treated as a fixed constant in the development of asymptotic results for the proposed SDA estimator. The selection of H must satisfy the coverage condition (C2) for estimating the central subspace $\mathcal{S}_{Y|\mathbf{X}}$. An empirical investigation on the impact of H is provided in Section 4.2.1.

3.2. Theoretical Properties

We first derive the necessary condition that $\hat{\mathbf{v}}_i$, the vector of $\hat{v}_{hi}$, can be an asymptotic linear estimator (ALE).

Lemma 1. If $\|\hat{\boldsymbol{\xi}}_i - \boldsymbol{\xi}_i^0\|_1 = o_p((\log p)^{-1/2})$ and $\log(p) = o(n)$, then we have

$$\sqrt{n}(\hat{\mathbf{v}}_i - \mathbf{v}_i^0) - \frac{1}{\sqrt{n}} \sum_{j=1}^n \boldsymbol{\psi}_i(\mathbf{W}_j) = o_p(1), \quad (7)$$

where $\boldsymbol{\psi}_i(\mathbf{W}_j) = (\psi_{1i}(\mathbf{W}_j), \dots, \psi_{Hi}(\mathbf{W}_j))^\top$ and $\psi_{hi}(\mathbf{W}) = I(Y \in \mathcal{I}_h)(X_i - \boldsymbol{\xi}_i^{0,\top} \mathbf{X}_{-i}) - v_{hi}^0$, where $\mathbf{W} = \{\mathbf{X}, Y\}$.

Lemma 1 requires that the l_1 -norm of $\hat{\boldsymbol{\xi}}_i - \boldsymbol{\xi}_i^0$ converges faster than the order of $(\log(p))^{-1/2}$ as $p \rightarrow \infty$. A proof of this lemma is provided in the supplemental material.

Due to the sparsity assumption of $\boldsymbol{\xi}_i^0$, we consider the LASSO estimator (Tibshirani 1996), which minimizes the following penalized least squares:

$$\hat{\boldsymbol{\xi}}_i = \arg \min_{\boldsymbol{\xi}_i \in \mathbb{R}^{p-1}} \frac{1}{2n} \|\mathbf{x}_i - \boldsymbol{\xi}_i^\top \mathbf{x}_{-i}\|_2^2 + \lambda_i \|\boldsymbol{\xi}_i\|_1, \quad (8)$$

where $\lambda_i \geq 0$ is the tuning parameter. Theoretical properties of LASSO estimators have been extensively studied in the literature (Greenshtein and Ritov 2004; Meinshausen and Bühlmann 2006; Bickel, Ritov, and Tsybakov 2009; Bühlmann and Van De Geer 2011). By further assuming the following regularity conditions on the design matrix $\mathbf{x}$:

- (C3) Restricted eigenvalue condition: Let $\mathcal{C}_a(\mathcal{I}_i) \subset \mathbb{R}^{p-1}$ be a set defined as $\mathcal{C}_a(\mathcal{I}_i) = \{\mathbf{b} \in \mathbb{R}^{p-1} : \|\mathbf{b}(\mathcal{I}_i^\top)\|_1 \leq a \|\mathbf{b}(\mathcal{I}_i)\|_1\}$, where $a > 0$. Then $\mathbf{x}_{-i}$ satisfies the restricted eigenvalue condition for $\kappa > 0$ if $n^{-1} \|\mathbf{b}^\top \mathbf{x}_{-i}\|_2^2 \geq \kappa \|\mathbf{b}\|_2^2$, for every $\mathbf{b} \in \mathcal{C}_a(\mathcal{I}_i)$.
- (C4) All columns of $\mathbf{x}$, denoted as $\mathbf{X}_l = (X_{l,1}, \dots, X_{l,n})^\top$ for $l = 1, \dots, p$, satisfy that $n^{-1} \|\mathbf{X}_l\|_2^2 \leq C$, where C is a positive constant.

we have the following bound for the l_1 -norm of $\hat{\boldsymbol{\xi}}_i - \boldsymbol{\xi}_i^0$:

Lemma 2. Under regularity conditions (C3) and (C4), when the tuning parameter λ_i satisfies

$$\lambda_i \geq \sqrt{\frac{24C\sigma_i^2 \log(p-1)}{n}}, \quad (9)$$

we have

$$\|\hat{\boldsymbol{\xi}}_i - \boldsymbol{\xi}_i^0\|_1 = O_p(\lambda_i I_i). \quad (10)$$

Lemma 2 is similar to the well-known result for the LASSO estimator (Bickel, Ritov, and Tsybakov 2009; Bühlmann and Van De Geer 2011). For the completion of our theoretical development, a proof of **Lemma 2** is provided in the supplemental material. The error bound (10) relies on the RE condition, which is commonly assumed in the literature (Bickel, Ritov, and Tsybakov 2009; Wainwright 2019). Intuitively, the condition regulates the Gram matrix $\mathbf{x}_{-i}^\top \mathbf{x}_{-i}/n$ so that the loss function would not be too flat at its minimizer. Although the RE condition is hard to verify in practice due to the unknown I_i , Raskutti, Wainwright, and Yu

(2010) proved that the RE condition holds with high probability for a broad class of Gaussian design matrices when the sample size satisfies $n = \Omega_p(I_i \log(p))$. The boundedness condition (C4) is also commonly assumed in the literature (Fan and Li 2001; Bühlmann and Van De Geer 2011). Under the Gaussian assumption for $\mathbf{X}$, one can show that this condition holds with a high probability when $\log(p) = o(n)$.

According to (10) and Lemma 1, we assume the following regularity conditions:

(C5) The number of predictors p satisfies $\lim_{n \rightarrow \infty} \log(p)/n \rightarrow 0$.

(C6) The tuning parameter λ_i satisfies (9) and $I_i = o(\sqrt{n}/\log(p))$.

Here, (C5) requires $p \ll e^n$ and (C6) imposes the sparsity condition. A direct corollary of Lemmas 1 and 2 shows that $\hat{\mathbf{v}}_i$, under regularity conditions, is an ALE when the LASSO estimator is used.

Corollary 1. Under regularity conditions (C3)–(C6), $\hat{\mathbf{v}}_i$ is an asymptotic linear estimator when $\hat{\boldsymbol{\xi}}_i$ is the LASSO estimator of $\boldsymbol{\xi}_i$.

We then study the asymptotic distribution of $n^{-1/2} \sum_{j=1}^n \boldsymbol{\psi}_i(\mathbf{w}_j)$. We first impose the following additional regularity conditions for the moments of $\mathbf{v}_i$:

(C7) Let $p_h = P(Y \in I_h)$, there exist positive constants $\gamma_1 \leq 1 \leq \gamma_2$ such that the probability p_h satisfies

$$\frac{\gamma_1}{H} \leq p_h \leq \frac{\gamma_2}{H} \text{ for every } h \in \mathbb{H}.$$

(C8) There exist positive constants γ_3 and γ_4 such that, for every $h \in \mathbb{H}$, we have

$$|E(Z_i|Y \in I_h)| < \gamma_3, E(Z_i^2|Y \in I_h) < \gamma_4.$$

Define $\boldsymbol{\Omega}_i = E[\boldsymbol{\psi}_i(\mathbf{W})\boldsymbol{\psi}_i^\top(\mathbf{W})]$, where $\Omega_i(h, h) = p_h E(Z_i^2|Y \in I_h) - p_h^2 [E(Z_i|Y \in I_h)]^2$ and $\Omega_i(h_1, h_2) = -p_{h_1} p_{h_2} E(Z_i|Y \in I_{h_1}) E(Z_i|Y \in I_{h_2})$, for every $h, h_1, h_2 \in \mathbb{H}$. We first derive an upper bound for the approximating error of the multivariate Gaussian distribution to the distribution of $n^{-1/2} \sum_{j=1}^n \boldsymbol{\psi}_i(\mathbf{w}_j)$.

Lemma 3. Under regularity conditions (C7) and (C8), if $\boldsymbol{\Omega}_i$ is invertible, there exists a constant C such that

$$\sup_{A \in \mathcal{C}_H} \left| P\left(\frac{1}{\sqrt{n}} \sum_{j=1}^n \boldsymbol{\psi}_i(\mathbf{w}_j) \in A\right) - P(N(0, \boldsymbol{\Omega}_i^0) \in A) \right| \leq C \frac{H^{1/4}}{n^{1/2}},$$

where $\mathcal{C}_H$ is defined as the set of all convex subsets of $\mathbb{R}^H$.

Lemma 3 is a direct consequence of the multidimensional Berry-Esseen Bound (Bentkus 2005), and a detailed proof is provided in the supplemental material. Finally, combining Corollary 1 and Lemma 3, we have proved the following theorem.

Theorem 1. Under regularity conditions (C3)–(C8), if $\boldsymbol{\Omega}_i$ is invertible, then $\sqrt{n}(\hat{\mathbf{v}}_i - \mathbf{v}_i^0) \xrightarrow{d} N(0, \boldsymbol{\Omega}_i^0)$.

In ultra-high dimensional settings, where $\log(p) = O(n^a)$ for some $a \in (0, 1 - 2\kappa)$ and $\kappa \geq 0$, the average in-sample prediction error of the LASSO estimator (8) will diverge as $n \rightarrow \infty$. Therefore, a sure independence screening (SIS) procedure, as proposed by Fan and Lv (2008), should be employed to reduce the dimensionality to a moderate scale, to satisfy condition (C4). Details of the SIS procedure are provided in the supplementary material.

Our asymptotic results are derived for inference on a single X_i . When conducting inference on a collection of indices $\mathbb{M} \subseteq \mathbb{P}$ with $|\mathbb{M}| = m \leq p$, the key step is to ensure that the high-probability event $\mathbb{E}_i$ defined in the supplementary material, Section A.3, (S4), can hold simultaneously over $i \in \mathbb{M}$. Notice that we set λ_i with the order as the standard theoretical λ used in LASSO estimators (Bickel, Ritov, and Tsybakov 2009), but with a larger constant in front to facilitate simultaneous control. Under this specification, we have shown, in the proof of Lemma 2, that $P(\mathbb{E}_i) \geq 1 - 2(p-1)^{-2}$. Directly applying the union bound gives $P(\bigcap_{i \in \mathbb{M}} \mathbb{E}_i) \geq 1 - 2m(p-1)^{-2} = 1 - o(1)$. Consequently, all intermediate steps in Lemmas 1–3, in particular, the ALE remainder control and the Berry-Esseen bound, can be generalized to results that uniformly apply for all $i \in \mathbb{M}$ by conditioning on $\bigcap_{i \in \mathbb{M}} \mathbb{E}_i$.

3.3. Standard Error Estimation

As shown in Theorem 1, the asymptotic variance of $\sqrt{n}(\hat{\mathbf{v}}_i - \mathbf{v}_i^0)$ is given by $\boldsymbol{\Omega}_i$, which is defined as $E[\boldsymbol{\psi}_i(\mathbf{W})\boldsymbol{\psi}_i^\top(\mathbf{W})]$. Denote $\hat{\boldsymbol{\psi}}_i(\mathbf{W}) = (\hat{\psi}_{1i}(\mathbf{W}), \dots, \hat{\psi}_{Hi}(\mathbf{W}))^\top$ to be the estimated influence vector, where $\hat{\psi}_{hi}(\mathbf{W}) = I(Y_i \in I_h)(X_i - \hat{\boldsymbol{\xi}}_i^\top \mathbf{X}_{-i}) - \hat{v}_{hi}$. We consider the following sample mean estimator:

$$\hat{\boldsymbol{\Omega}}_i = \frac{1}{n} \sum_{j=1}^n \hat{\boldsymbol{\psi}}_i(\mathbf{W}_j) \hat{\boldsymbol{\psi}}_i^\top(\mathbf{W}_j).$$

The consistency of $\hat{\boldsymbol{\Omega}}_i$ is given in Theorem 2, and a detailed proof is provided in the supplemental material.

Theorem 2. Under regularity conditions (C3)–(C8), we have $\hat{\boldsymbol{\Omega}}_i \xrightarrow{P} \boldsymbol{\Omega}_i^0$.

3.4. Hypothesis Testing

3.4.1. Chi-Squared Test

According to Proposition 1, the conditional independence test (2) is equivalent to

$$H_0 : \mathbf{v}_i = \mathbf{0} \text{ versus } H_1 : \mathbf{v}_i \neq \mathbf{0}. \quad (11)$$

With asymptotic results in Sections 3.2 and 3.3, consider the Wald Chi-squared test statistic $\hat{T}_i^\chi = \hat{\mathbf{v}}_i^\top (\hat{\boldsymbol{\Omega}}_i/n)^{-1} \hat{\mathbf{v}}_i$. The following corollary provides the asymptotic distribution of $\hat{T}_i^\chi$.

Corollary 2. Under regularity conditions (C1)–(C8),

$$\hat{T}_i^\chi \xrightarrow{d} \chi_H^2(v_i^{0,\top} \boldsymbol{\Omega}_i^{-1} v_i^0),$$

where $\chi_H^2(\lambda)$ is the noncentral Chi-squared distribution with H degrees of freedom and noncentral parameter λ .

Corollary 2 is an immediate consequence of **Theorems 1** and **2**. Under H_0 , the asymptotic distribution of $\hat{T}_i^\chi$ is a central chi-squared distribution with H degrees of freedom. We denote the hypothesis testing approach based on the Chi-squared statistic by SDA- χ^2 .

3.4.2. Kolmogorov-Smirnov and Cramér-von-Mises Tests

Alternatively, because the test of hypothesis (11) is equivalent to $H_0 : v_{hi} = 0$, for all $h \in \mathcal{H}$, versus $H_1 : v_{hi} \neq 0$, for some $h \in \mathcal{H}$, we can consider each univariate test and then combine. The asymptotic normality in **Theorem 1** suggests a test statistic $z_{hi} = \sqrt{n}\hat{v}_{hi}/\sqrt{\hat{\Omega}_i(h, h)}$ and $z_{hi} \xrightarrow{d} N(0, 1)$ when $v_{hi} = 0$. Here, we consider a Kolmogorov-Smirnov (KS) type statistic:

$$\hat{T}_i^{\text{KS}} = \max_{h \in \mathcal{H}} |z_{hi}| = \max_{h \in \mathcal{H}} \left| \frac{\sqrt{n}\hat{v}_{hi}}{\sqrt{\hat{\Omega}_i(h, h)}} \right|. \quad (12)$$

Thus, we reject H_0 if $\hat{T}_i^{\text{KS}} > c_i$ where c_i is some critical value. We also develop a Cramér-von-Mises (CvM) type statistic $\hat{T}_i^{\text{CvM}} = \int |z_{hi}| dQ(h)$, where Q is a weight function on $\mathcal{H}$. When using equal weights, the CvM-type statistic is defined as

$$\hat{T}_i^{\text{CvM}} = \frac{1}{H} \sum_{h=1}^H |z_{hi}| = \frac{1}{H} \sum_{h=1}^H \left| \frac{\sqrt{n}\hat{v}_{hi}}{\sqrt{\hat{\Omega}_i(h, h)}} \right|. \quad (13)$$

We name the proposed tests based on (12) and (13) as SDA-KS and SDA-CvM, respectively.

Because the asymptotic distributions of $\hat{T}_i^{\text{KS}}$ and $\hat{T}_i^{\text{CvM}}$ are difficult to derive analytically, we adopt a simulation-based approach referred to as the multiplier bootstrap (MB) (van der Vaart and Wellner 2000; Chernozhukov, Chetverikov, and Kato 2013). The theoretical development of this method and a pseudocode are provided in the supplemental material.

3.5. Multiple Hypothesis Testing

We propose an approach similar to the knockoff filter (Barber and Candès 2015; Candès et al. 2018) for multiple hypothesis testing with FDR control. To generate a knockoff copy of $\mathbf{X}$, denoted as $\tilde{\mathbf{X}}$, there are two requirements: (a) each $\tilde{X}_i$ in $\tilde{\mathbf{X}}$ is exchangeable with the corresponding X_i in $\mathbf{X}$; and (b) $\tilde{\mathbf{X}}$ provides no further regression information about the response Y , that is, $\mathbf{X}$ and $\tilde{\mathbf{X}}$ need to be as dissimilar as possible. Generating knockoff copies can be challenging as it requires the distribution of $\mathbf{X}$ to be completely known (Candès et al. 2018) or can be consistently estimated (Barber, Candès, and Samworth 2020). However, our proposed SDA has an advantage due to the standardized variable Z_i . Thus, we can generate $\tilde{z}_i$, a random sample of size n from $N(0, \hat{\sigma}_i^2)$, which will suffice.

Let $\tilde{T}_i$ be the test statistic calculated using the knockoff copy $\tilde{z}_i$, we obtain the feature statistic $M_i = M(\hat{T}_i, \tilde{T}_i)$, where $M(\cdot, \cdot)$ is an antisymmetric function. Following **Corollary 1**, the distribution of M_i is asymptotically symmetric to 0 for $i \in \mathcal{A}^c$ (more details are available in the proof of **Theorem 3**). Therefore, with a sufficient number of hypotheses, given a threshold value t , $\#\{i : M_i \leq -t\}$ can be used as a conservative estimate of the

number of false selections, regardless of the exact distribution of M_i under the null. Thus, with a desired FDR level q , the set of selected variables is defined as $\{i : M_i \geq \tau\}$, where the data-dependent threshold τ is defined as

$$\tau = \min \left\{ t : \frac{\#\{i : M_i \leq -t\}}{\#\{i : M_i \geq t\}} \leq q \right\}. \quad (14)$$

The pseudocode for the implementation of the proposed multiple hypothesis testing procedure is summarized in **Algorithm 1**. The next theorem, with a detailed proof available in the supplementary material, shows that both false positive proportion (FDP), which is defined as the proportion of false selections among all selected variables, and FDR can be controlled asymptotically using the threshold τ defined in (14).

Theorem 3. Under (C1)-(C8), let $p_0 = |\mathcal{A}^c|$, as $n \rightarrow \infty$ and $p_0 \rightarrow \infty$, the SDA procedure in **Algorithm 1** satisfies

$$P(\text{FDP}(\tau) \leq q) \rightarrow 1 \text{ and } \limsup_{p_0 \rightarrow \infty} \text{FDR}(\tau) \leq q.$$

Algorithm 1 False discovery rate control via SDA.

- 1: Divide the range of Y into H slices
 - 2: **for** $i = 1, \dots, p$ **do**
 - 3: Obtain $\hat{z}_i$ and $\hat{\sigma}_i^2$ by fitting a high-dimension regression model $X_i = \boldsymbol{\zeta}_i^\top \mathbf{X}_{-i} + Z_i$
 - 4: Generate a knockoff copy $\tilde{z}_i$ by randomly drawing n sample from $N(0, \hat{\sigma}_i^2)$
 - 5: Calculate the test statistic $\hat{T}_i$ and $\tilde{T}_i$ from $\hat{z}_i$ and $\tilde{z}_i$, respectively
 - 6: Calculate the feature statistic M_i
 - 7: **end for**
 - 8: Calculate the data-dependent threshold τ defined in (14)
 - 9: **for** $i = 1, \dots, p$ **do**
 - 10: **if** $M_i > \tau$ **then**
 - 11: Reject H_i
 - 12: **else**
 - 13: Do not reject H_i
 - 14: **end if**
 - 15: **end for**
-

4. Simulation Studies

In this section, we investigate the empirical performance of the proposed SDA- χ^2 , SDA-KS, and SDA-CvM procedures through extensive simulation scenarios.

4.1. Simulation Settings

We set the significance level at 0.05 and FDR at 0.1 for multiple hypothesis testing. We consider $n = 200$ or 400 , and $p = 1000$ or 2000 . Tuning parameters of $\hat{\zeta}_i$ for all methods are selected based on 10-fold cross-validation.

4.1.1. Correlation Structures

Fixed precision matrix. We first consider the setting with fixed correlation structures. We generate covariates $\mathbf{X}$ from a multivariate Gaussian distribution with mean zero and precision

matrix Θ . Here, we let Θ be a block diagonal matrix with cluster sizes of $q = 5$. Within each block, we let $\Theta_{ii} = 1$ and $\Theta_{i'i'} = 0.5$ for $i \neq i'$. We denote $\mathcal{B} = \{1, \dots, B\}$, where $B = p/q$, the index set of blocks.

Random network covariates. We then consider the scenario where the correlation structure of $\mathbf{X}$ is determined by a randomly generated small-world network (Watts and Strogatz 1998). Specifically, each X_i is connected to covariates within $e = 5$ neighbors, with a rewiring probability of 0.25. For each connected pair $(X_i, X_{i'})$, the corresponding entry $\theta_{i'i'}$ in the precision matrix is uniformly sampled from $(-1, -0.5) \cup (0.5, 1)$.

4.1.2. Regression Functions

We consider the following two single-index models (1 and 2) and two multiple-index models (3 and 4): Model 1: $Y = \mathbf{b}^\top \mathbf{X} + \epsilon$; Model 2: $Y = \sin(\mathbf{b}^\top \mathbf{X}) \exp(\mathbf{b}^\top \mathbf{X}) + \epsilon$; Model 3: $Y = 3\mathbf{b}_1^\top \mathbf{X} / [0.5 + (1.5\mathbf{b}_2^\top \mathbf{X})^2] + \epsilon$; Model 4: $Y = \sum_{k=1}^d (0 \vee \mathbf{b}_k^\top \mathbf{X}) + \epsilon$, where $\epsilon \sim N(0, 1)$ and $\mathbf{X} \perp \epsilon$.

4.2. Simulation Results

We discuss key findings of simulation studies from multiple perspectives in the following five sub-sections. Detailed simulation settings are included in the supplemental material.

4.2.1. The Choice of H

We first investigate the impact of the choice of H . Results for SDA-CvM, SDA- χ^2 , and SDA-KS are presented in Figures 1, S1, and S2, respectively. Across all regression models, the empirical Type I error rates for the null variables are nearly unaffected by the choice of H . In the two single-index models (models 1 and 2), the empirical power for $\mathbf{b}_1$ (the larger effect size) is consistently 1 across all values of H . For $\mathbf{b}_2$ (the smaller effect size), the power decreases with increasing H when $n = 200$, but this decreasing trend is less pronounced when $n = 400$. In model

3, the two active variables in $\mathbf{b}_1$ show patterns similar to those in models 1 and 2, while the two active variables in $\mathbf{b}_2$ behave differently. Specifically, the empirical power for X_3 (larger effect size) increases sharply from the null level at $H = 2$ and then stabilizes, whereas for X_4 (smaller effect size), the power increases and then gradually decreases. In model 4, although it is a multiple-index nonlinear model, the regression function is close to a linear one, leading to a decreasing trend in power across all X_i as H increases.

In summary, the optimal choice of H depends on the sample size, the effect size, and the form of regression functions. Values of H between 4 and 7 provide robust performance across all settings considered in this study. Therefore, we set $H = 5$ for the remaining simulation studies.

4.2.2. Empirical Selection Rates

We then study and compare the empirical Type I error rates and power for SDA- χ^2 , SDA-KS, and SDA-CvM. To compare with existing methods, we also include the selective inference (SI) method (Lee et al. 2016; Taylor and Tibshirani 2018) and the high-dimensional permutation (HP) test based on the partial correlation (Hemerik, Thoresen, and Finos 2021). Empirical selection rates for the first 100 covariates are calculated based on 1000 simulated datasets.

The SI method performs poorly in nonlinear settings due to very low selection rates of active variables under the LASSO estimator, as shown in Table S1. Since SI only applies to variables selected by LASSO, the low selection rates also result in low empirical power. Therefore, we focus our comparison on the proposed SDA methods and the HP method. Tables 1 and S2 summarize the simulation results for the fixed and network precision matrix structures, respectively. All methods conservatively control the Type I error across all settings, with the exception of SDA-CvM and SDA- χ^2 under the fixed precision matrix in model 1. The empirical power of all methods increases

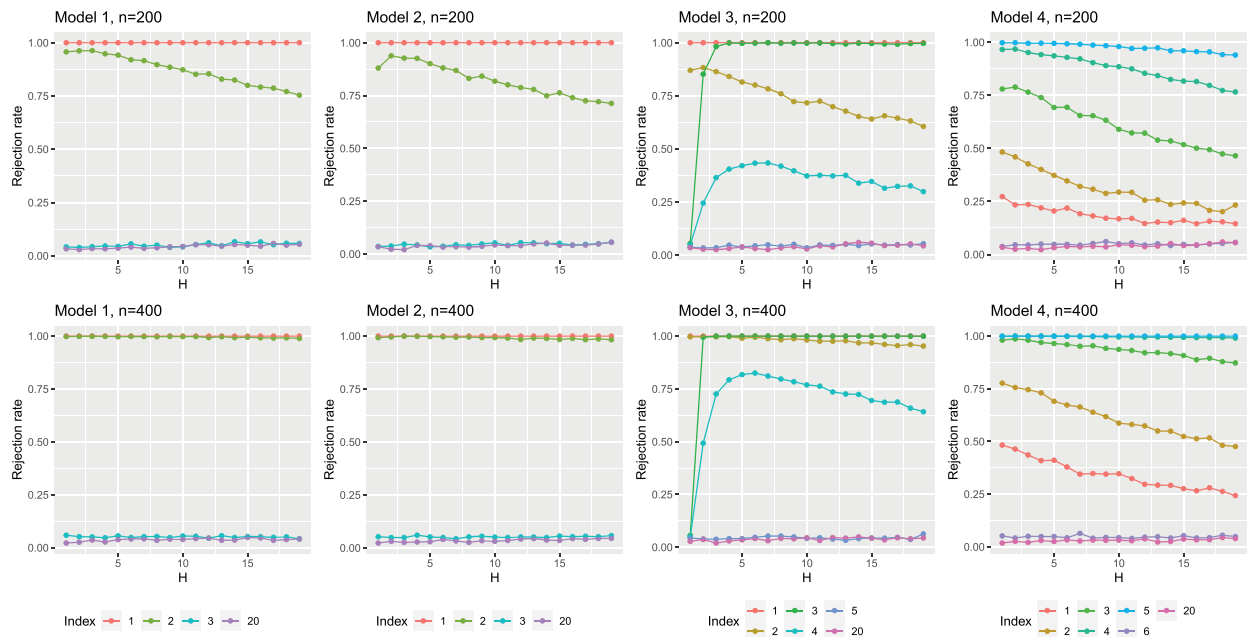


Figure 1. Empirical Type I error rates (for two selected null covariates) and power (for all active signals) with respect to H for SDA-CvM.

Table 1. Empirical power and Type I error rates for fixed precision matrix.

n	p	Method	X_1	X_2	X_6	X_7	X_{11}	X_{12}	X_{16}	X_{17}	Null
Model 1											
400	1000	SDA-KS	0.621	***	0.979	***	0.999	1.000	1.000	1.000	0.030
		SDA-CvM	0.719	***	0.997	***	1.000	1.000	1.000	1.000	0.044
		SDA- χ^2	0.781	***	0.998	***	1.000	1.000	1.000	1.000	0.042
		HP	0.514	***	0.644	***	0.675	0.670	0.714	0.724	0.073
200	1000	SDA-KS	0.301	***	0.750	***	0.889	0.868	0.993	0.996	0.030
		SDA-CvM	0.417	***	0.826	***	0.951	0.932	0.998	0.999	0.047
		SDA- χ^2	0.429	***	0.852	***	0.963	0.950	1.000	1.000	0.047
		HP	0.126	***	0.300	***	0.385	0.387	0.558	0.556	0.026
400	2000	SDA-KS	0.634	***	0.981	***	1.000	0.998	1.000	1.000	0.036
		SDA-CvM	0.751	***	0.988	***	1.000	1.000	1.000	1.000	0.052
		SDA- χ^2	0.794	***	0.995	***	1.000	1.000	1.000	1.000	0.051
		HP	0.256	***	0.562	***	0.624	0.612	0.706	0.707	0.037
200	2000	SDA-KS	0.310	***	0.764	***	0.884	0.881	0.994	0.991	0.034
		SDA-CvM	0.432	***	0.862	***	0.939	0.931	0.998	0.997	0.055
		SDA- χ^2	0.447	***	0.890	***	0.952	0.948	1.000	0.998	0.056
		HP	0.076	***	0.144	***	0.172	0.184	0.269	0.258	0.022
Model 2											
400	1000	SDA-KS	0.218	***	0.598	***	0.803	0.819	0.980	0.976	0.025
		SDA-CvM	0.300	***	0.739	***	0.905	0.913	0.995	0.986	0.037
		SDA- χ^2	0.318	***	0.764	***	0.922	0.925	0.997	0.993	0.036
		HP	0.508	***	0.666	***	0.674	0.681	0.729	0.751	0.074
200	1000	SDA-KS	0.103	***	0.289	***	0.388	0.389	0.681	0.689	0.022
		SDA-CvM	0.163	***	0.379	***	0.508	0.515	0.772	0.762	0.036
		SDA- χ^2	0.166	***	0.396	***	0.519	0.528	0.813	0.807	0.035
		HP	0.107	***	0.328	***	0.408	0.387	0.568	0.575	0.027
400	2000	SDA-KS	0.214	***	0.620	***	0.802	0.816	0.976	0.968	0.026
		SDA-CvM	0.334	***	0.776	***	0.901	0.902	0.996	0.991	0.037
		SDA- χ^2	0.343	***	0.791	***	0.913	0.913	0.998	0.992	0.037
		HP	0.277	***	0.557	***	0.620	0.634	0.720	0.733	0.038
200	2000	SDA-KS	0.108	***	0.295	***	0.408	0.415	0.665	0.667	0.025
		SDA-CvM	0.170	***	0.394	***	0.518	0.533	0.775	0.779	0.038
		SDA- χ^2	0.163	***	0.413	***	0.533	0.555	0.806	0.803	0.039
		HP	0.065	***	0.154	***	0.182	0.171	0.254	0.272	0.022
Model 3											
400	1000	SDA-KS	0.926	0.818	1.000	0.404	1.000	0.967	***	***	0.021
		SDA-CvM	0.954	0.943	1.000	0.520	1.000	0.997	***	***	0.030
		SDA- χ^2	0.970	0.939	1.000	0.553	1.000	0.997	***	***	0.028
		HP	0.717	0.024	0.960	0.045	1.000	0.516	***	***	0.034
200	1000	SDA-KS	0.649	0.368	1.000	0.153	0.987	0.658	***	***	0.017
		SDA-CvM	0.753	0.628	1.000	0.228	1.000	0.853	***	***	0.029
		SDA- χ^2	0.784	0.611	1.000	0.227	0.999	0.845	***	***	0.028
		HP	0.297	0.007	0.681	0.013	0.822	0.444	***	***	0.021
400	2000	SDA-KS	0.925	0.818	1.000	0.396	1.000	0.984	***	***	0.021
		SDA-CvM	0.967	0.944	1.000	0.547	1.000	0.998	***	***	0.031
		SDA- χ^2	0.977	0.938	1.000	0.555	1.000	0.999	***	***	0.030
		HP	0.543	0.010	0.863	0.043	0.978	0.511	***	***	0.028
200	2000	SDA-KS	0.690	0.393	1.000	0.211	0.994	0.687	***	***	0.019
		SDA-CvM	0.788	0.615	1.000	0.276	0.999	0.879	***	***	0.032
		SDA-CvM	0.819	0.605	1.000	0.284	0.999	0.866	***	***	0.031
		HP	0.228	0.004	0.442	0.024	0.516	0.248	***	***	0.015
Model 4											
400	1000	SDA-KS	0.414	***	0.899	***	0.999	1.000	0.941	0.989	0.020
		SDA-CvM	0.459	***	0.940	***	1.000	1.000	0.977	0.994	0.029
		SDA- χ^2	0.518	***	0.961	***	1.000	1.000	0.984	0.996	0.028
		HP	0.696	***	0.946	***	1.000	0.519	0.496	0.500	0.012
200	1000	SDA-KS	0.200	***	0.586	***	0.942	0.994	0.700	0.870	0.017
		SDA-CvM	0.263	***	0.680	***	0.977	0.998	0.825	0.940	0.028
		SDA- χ^2	0.278	***	0.723	***	0.990	0.998	0.826	0.948	0.027
		HP	0.305	***	0.635	***	0.803	0.442	0.287	0.378	0.007
400	2000	SDA-KS	0.412	***	0.901	***	0.997	1.000	0.950	0.997	0.020
		SDA-CvM	0.472	***	0.929	***	0.999	1.000	0.979	1.000	0.029
		SDA- χ^2	0.529	***	0.962	***	0.999	1.000	0.990	1.000	0.028
		HP	0.514	***	0.858	***	0.980	0.502	0.448	0.478	0.008
200	2000	SDA-KS	0.217	***	0.618	***	0.945	0.991	0.744	0.885	0.018
		SDA-CvM	0.297	***	0.691	***	0.989	1.000	0.838	0.953	0.029
		SDA- χ^2	0.312	***	0.729	***	0.993	1.000	0.856	0.958	0.026
		HP	0.230	***	0.445	***	0.514	0.250	0.146	0.169	0.009

with larger sample sizes, stronger effect sizes, or larger cluster sizes. Among the three SDA statistics, SDA-CvM and SDA- χ^2 perform similarly and consistently exhibit higher power than SDA-KS. In all settings, the SDA-based methods outperform the HP method.

4.2.3. Multiple Hypothesis Testing

In this section, we study the performance of the proposed multiple hypothesis testing procedure introduced in Section 3.5. Based on the simulation results in Section 4.2.2, which show that SDA-CvM and SDA- χ^2 outperform SDA-KS, we focus on SDA-CvM as the test statistic. We consider two feature statistics: coefficient difference and sign-max, referred to as CvMCD-SDA and CvMSM-SDA, respectively. As a benchmark method, we also include the model-X knockoff procedure proposed by Candès et al. (2018), using the LASSO coefficient-difference statistic (LCD-Knockoff).

Histograms of the FDP and power, based on 200 simulated datasets, are shown in Figure S3. Here, power is defined as the proportion of active variables correctly selected. Both CvMCD-SDA and CvMSM-SDA control the FDR (the expected value of FDP) at the nominal 0.1 level, with CvMCD-SDA being more conservative. CvMSM-SDA also achieves higher power across all settings. When compared with the LCD-Knockoff method, our proposed procedures perform better in models 2 and 3 but slightly worse in model 1 (linear model). Notably, LCD-Knockoff performs the worst in model 2, where in more than 70% of the simulations, no variable is selected.

4.2.4. Impact of the Covariates Distribution

Our proposed method relies on the normality assumption for the covariates X . In this section, we evaluate its robustness under alternative distributions of X . We consider three multivariate

t -distributions and one multivariate Chi-squared distribution, with detailed settings available in the supplemental material.

Table 2 summarizes the simulation results for empirical selection rates. Our method successfully controls the Type I error for T_5^{MVT} , T_3^{MVT} , and T_5^{GC} in all settings, with a slight inflation observed under $\chi_5^{2,GC}$ in models 1 and 2. In terms of power, the method performs only slightly worse than the multivariate Gaussian case (Table S2) under T_5^{MVT} and T_3^{MVT} , and similarly under T_5^{GC} . The results indicate that our method is robust under the elliptical family. For $\chi_5^{2,GC}$, its performance is similar to the Gaussian case in models 1 and 2, but lower power is observed for X_7 in model 3 and for X_1 and X_{11} in model 4.

Figure S4 presents histograms of FDP and power. Although the FDR (i.e., the average FDP) is controlled at the nominal level for both distributions, the FDP exhibits substantially greater variability in the non-Gaussian case, with a noticeably larger fraction of outliers than in the corresponding Gaussian setting. In addition, the average power is lower and the power distribution is more dispersed than in the Gaussian case.

4.2.5. Impact of the Precision Matrix Sparsity

In this section, we investigate the impact of the precision matrix sparsity. We focus on the fixed precision matrix (block diagonal) specified in Section 4.1, considering two sparsity levels: $q = 5$ and $q = 10$. We examine three estimators of ξ_i : (a) the LASSO estimator, denoted by $\hat{\xi}_i$; (b) the LASSO estimator with a preliminary SIS step that reduces dimensionality to $\lfloor n/\log(n) \rfloor$, denoted by $\hat{\xi}_i^{SIS}$; and (c) the “oracle” estimator, which assumes that the active set $\mathcal{I}_i$ is known and applies least squares estimation conditional on $\mathcal{I}_i$, denoted by $\hat{\xi}_i^{OR}$.

Figures 2 and S5 summarize the simulation results for the variables across the five blocks. For $\hat{\xi}_i$, the empirical Type I error rate increases and power decreases as the number of active

Table 2. Empirical power and Type I error rates with respect to different distributions of X .

Dist.	X_1	X_2	X_6	X_7	X_{11}	X_{12}	X_{16}	X_{17}	Null
Model 1									
T_5^{MVT}	0.725	***	0.992	***	1.000	0.999	1.000	1.000	0.040
T_3^{MVT}	0.681	***	0.986	***	0.998	1.000	1.000	1.000	0.045
T_5^{GC}	0.787	***	0.999	***	1.000	1.000	1.000	1.000	0.044
$\chi_5^{2,GC}$	0.873	***	1.000	***	1.000	1.000	1.000	1.000	0.064
Model 2									
T_5^{MVT}	0.353	***	0.779	***	0.928	0.933	0.993	0.994	0.036
T_3^{MVT}	0.326	***	0.750	***	0.897	0.907	0.993	0.991	0.034
T_5^{GC}	0.362	***	0.796	***	0.927	0.922	0.992	0.994	0.035
$\chi_5^{2,GC}$	0.752	***	0.996	***	1.000	0.999	1.000	1.000	0.055
Model 3									
T_5^{MVT}	0.957	0.847	1.000	0.373	1.000	0.968	***	***	0.030
T_3^{MVT}	0.926	0.751	1.000	0.287	1.000	0.923	***	***	0.030
T_5^{GC}	0.979	0.903	1.000	0.581	1.000	0.980	***	***	0.034
$\chi_5^{2,GC}$	0.964	0.791	1.000	0.139	1.000	0.816	***	***	0.037
Model 4									
T_5^{MVT}	0.485	***	0.927	***	0.999	1.000	0.939	0.993	0.027
T_3^{MVT}	0.467	***	0.878	***	0.992	0.999	0.881	0.974	0.028
T_5^{GC}	0.542	***	0.954	***	1.000	1.000	0.978	0.995	0.027
$\chi_5^{2,GC}$	0.025	***	1.000	***	0.115	1.000	0.896	1.000	0.040

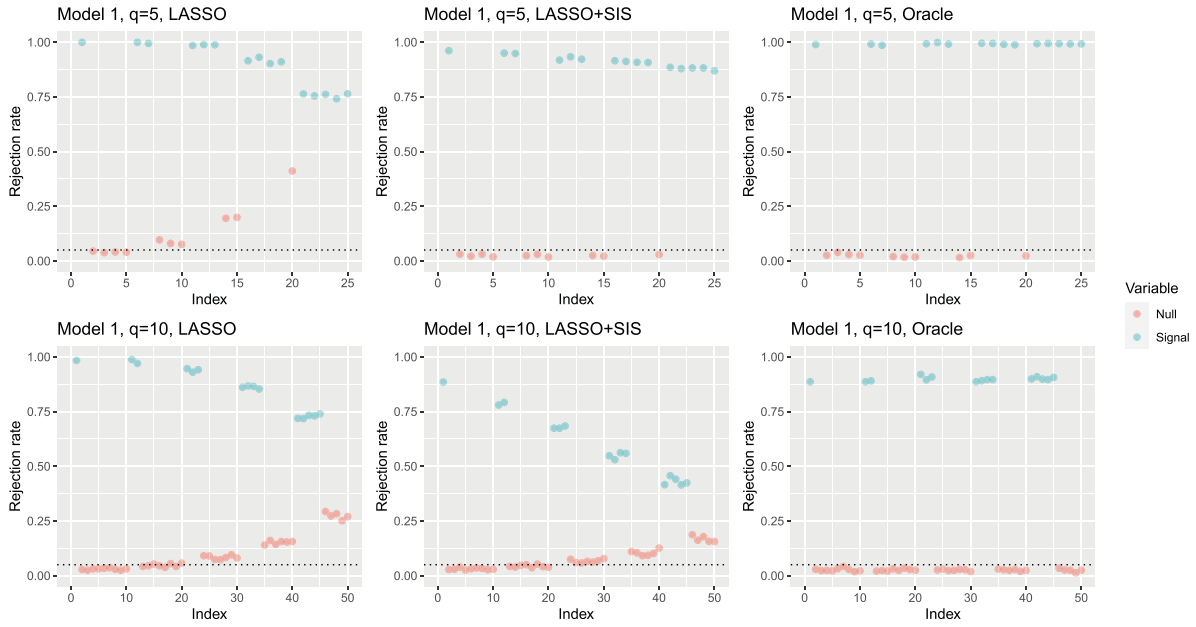


Figure 2. Empirical Type I error rates and power for different sparsity levels.

variables within the same block increases. When $q = 5$, incorporating the SIS step helps control the inflated Type I error rate. When $q = 10$, the Type I error becomes less inflated, but the power drops more sharply for $\hat{\xi}_i^{\text{SIS}}$. The oracle estimator $\hat{\xi}_i^{\text{OR}}$ maintains stable Type I error and power across all settings.

This simulation study highlights the limitations of the LASSO estimator in high-dimensional and non-sparse settings, where it fails to fully account for the influence of other active variables that are correlated with the target variable. Incorporating an SIS step can mitigate this issue when the sparsity level is moderate. However, as sparsity decreases further, the SIS step alone becomes insufficient, and then an alternative estimation strategy, such as SCAD or adaptive LASSO, may be considered.

5. Gene Expressions Associated with Alzheimer's Disease

The Alzheimer's Disease Neuroimaging Initiative (ADNI) study, launched in 2003 and led by Principal Investigator Dr. Michael W. Weiner, was established to support the development of treatments for Alzheimer's disease (AD) by tracking disease-related biomarkers, including serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment, over time. For up-to-date information, see www.adni-info.org. This longitudinal, multi-center study collected a comprehensive set of clinical, imaging, and genetic data from participants aged 55–90 across the United States and Canada. Participants included individuals with normal aging, mild cognitive impairment, dementia, and AD. Among other clinical variables, the ADNI dataset includes the Mini-Mental State Examination (MMSE) (Folstein, Folstein, and McHugh 1975), a widely used screening tool for cognitive function. While the optimal MMSE cutoff for identifying cognitive impairment remains a topic of debate (Chapma et al. 2016; Salis, Costaggu, and Mandas 2023), a commonly used threshold for the diagnosis of dementia is a

score of 24 or below, out of a maximum score of 30 (Tombaugh and McIntyre 1992; Zhang et al. 2021).

In this study, we apply the proposed SDA method to the ADNI microarray gene expression data to identify genes associated with MMSE scores. The gene expression data were obtained from the ADNI database (adni.loni.usc.edu, downloaded on April 27, 2024). Although the dataset includes microarray data for 745 individuals, we restrict our analysis to the 292 individuals with both gene expression and MMSE available at the same study visit.

Due to the ultra-high dimensionality, we perform an initial variable screening, specifically the SIS (Fan and Lv 2008), using Spearman's ρ correlation coefficients. The sure screening property of the marginal correlation statistic guarantees that the procedure has a high probability of including all the relevant variables, under regularity conditions. We pre-select $p = 2000$ probes, a relatively large number, to ensure that no relevant probes will be excluded from the screening. Note that, although we pre-select 2000 probes, their conditional associations with the MMSE score will be evaluated given all the remaining 49,385 probes.

Guided by our simulation findings in Section 4.2.3, we implement Algorithm 1 using the CvMSM-SDA test statistic on the 2000 selected probes. To account for the ultra-high dimensionality when conditioning on the remaining probes, we apply the SIS procedure described in the supplementary material with $\gamma = n/\log(n)$.

The study reveals that, at FDR of 0.1, the CvMSM-SDA selects four probes. We compare this result with existing literature and find that all four selected probes are known to be expressed more highly in AD patients than in normal patients. At a more liberal FDR of 0.2, our method identifies an additional seven probes. Among these probes, the targets of six probes have identified associations with AD in the literature, and the extra probe is a new finding. All of the identified probes, with literature references, are summarized in Table S3.

6. Discussion

This article explores high-dimensional statistical inference leveraging the theory of sufficient dimension reduction. Specifically, we propose the sufficient dimension association, a model-free measure for the conditional dependence between each predictor and the response variable. We prove that, under regularity conditions, the asymptotic normality for the proposed SDA estimator can be achieved in high-dimensional settings when $\log(p) = o(n)$. Based on the central limit theorem proven in this article, we construct SDA- χ^2 , SDA-KS, and SDA-CvM test statistics along with a multiplier bootstrap algorithm for a single test.

We also develop a knockoff-SDA method for multiple hypothesis testing with FDR control. One advantage for the proposed method is that the SDA statistics, as well as the corresponding knockoffs, can be obtained separately for each variable, which is easy for parallel computations and memory-efficient. Furthermore, our SDA procedure does not require estimating the distribution of $\mathbf{X}$, which is crucial for large-scale studies. For ultra-high dimensional data, such as the ADNI gene expression dataset, estimating a large but sparse covariance or precision matrix can be computationally challenging, due to the requirement of huge memory space to store the large matrix and extensive computations associated with it.

The validity of the proposed method relies on the normality of $\mathbf{X}$ and the sparsity of Θ . Such conditions are commonly assumed when analyzing gene expression data. Normality of the gene expression data can be assumed either after normalization, as in the case of microarray data, or after both the mean-variance modeling and normalization, as in the case of RNAseq read-counts (Law et al. 2014). Furthermore, our simulation results demonstrate that our proposed method is robust against model misspecification, especially within the elliptical family. Sparsity of gene regulatory networks is a common assumption in statistical and computational biology methods development (Noor et al. 2012; Wang et al. 2024). While a gene network as a whole may be highly complex, each gene is expected to interact strongly with only a few other members of the network. The covariance matrix of the gene expression levels is thus assumed to be sparse, and its inverse can be estimated using sparsity-based methods such as the graphical lasso (Wang et al. 2024).

Conservative Type I error rates. Our simulation studies indicate that the proposed method tends to produce conservative Type I error rates. This issue is likely due to the use of SIR for constructing the sequence of transformation functions, where using indicator functions to capture local conditional dependence between Y and Z_i may result in information loss, particularly favoring the null hypothesis. In future work, we may explore alternative approaches, such as splines or polynomial functions, for constructing the set of transformation functions $g_h(\cdot)$ to address this limitation.

Survival outcome. This work, motivated by the ADNI study, focuses on continuous outcomes. The proposed method can be easily applied to survival outcomes. Let T denote the survival time, C denote the censoring time, and $\Delta = I(C > T)$. Our outcome variable becomes (Y, Δ) , where $Y = T(1 - \Delta) + C\Delta$. By assuming $(T, C) \perp\!\!\!\perp \mathbf{X} | \mathbf{B}^T \mathbf{X}$, Cook (2003) shows that $\mathcal{S}_{(Y, \Delta) | \mathbf{X}} \subseteq \mathcal{S}_{T | \mathbf{X}}$ and the sufficient predictors for the regression

$(Y, \Delta) | \mathbf{X}$ are also sufficient predictors for the regression $T | \mathbf{X}$. By slicing the bivariate outcome (Y, Δ) , we can stratify Y based on the censoring indicator Δ and separately partition the range of Y into $H_{\Delta=0}$ and $H_{\Delta=1}$ slices, with $H_{\Delta=0} + H_{\Delta=1} = H$. Without loss of generality, we assume balanced slices so that $n = cH$, where $|J_1| = \dots = |J_H| = c$.

Network information. Our simulation study in Section 4.2.5 suggests that the LASSO estimator may lead to inflated Type I error rates and reduced power when the precision matrix is non-sparse. This issue can be addressed by using the least squares estimator, provided that the correlation structure of $\mathbf{X}$ is known. As noted by Li and Li (2008), network information is often available in gene expression studies, and genes connected within a network tend to exhibit similar regression coefficients. This motivates a potential future direction: incorporating network information into the proposed method. Specifically, we can modify the formulation in (8) by including only the variables that are connected to X_i within the network, and replacing the l_1 -penalty with the l_2 -penalty. Furthermore, methods for jointly testing the significance of groups of variables based on network connections can also be developed.

Supplemental Materials

In the supplemental material, we provide proofs of theorems, additional simulation results and discussions referenced in Section 4, additional results referenced in Section 5, and detailed funding information of ADNI study. The file “SDA.R” contains the R functions and demonstration code for the proposed method, and the complete code required to reproduce all simulation results in the article is available on <https://github.com/shyye008/SDA>.

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Disclosure Statement

The authors report there are no competing interests to declare.

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